Supplementary Information to:

Animal Studies for the Evaluation of *In Situ* Tissue-Engineered Vascular Grafts - a Systematic Review, Evidence Map and Meta-Analysis

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Supplementary Table 1: Search strategy for Pubmed using search string defining the components “blood vessel prosthesis”, “tissue engineering” and “animal”

Component	Search entry
Component 1: blood vessel prosthesis	“Blood Vessel Prosthesis”[MeSH] OR (((artery [tiab] OR arteries [tiab] OR vein [tiab] OR veins [tiab] OR vascular [tiab] OR arterial [tiab] OR venous [tiab]) AND (prostheses [tiab] OR prosthesis [tiab] OR graft [tiab] OR grafts [tiab] OR scaffold [tiab] OR scaffolds [tiab] OR replacement [tiab] OR replacements [tiab] OR substitute [tiab] OR substitutes [tiab] OR grafting [tiab])
Component 2: tissue engineering	AND (“tissue engineering” [tiab] OR “tissue engineered” [tiab] OR “Tissue Engineering” [MeSH Terms] OR “Guided Tissue Regeneration” [MeSH] OR regeneration [tiab]))) OR “TEVG”[tiab] OR ((artery [tiab] OR arteries [tiab] OR vein [tiab] OR veins [tiab] OR vascular [tiab] OR arterial [tiab] OR venous [tiab]) AND (“tissue engineering” [tiab] OR “tissue engineered” [tiab] OR “Tissue Engineering” [MeSH Terms] OR “Guided Tissue Regeneration” [MeSH])) OR “Tissue engineered blood vessel” [tiab] OR “Tissue engineered blood vessels” [tiab] OR “Blood vessel tissue engineering” [tiab] OR “Blood vessel regeneration” [tiab] OR (((degradable[tiab] OR biodegradable[tiab] OR resorbable[tiab] OR bioresorbable[tiab] OR regenerative[tiab]) AND (“blood vessel”[tiab] OR artery [tiab] OR arteries [tiab] OR vein [tiab] OR veins [tiab] OR vascular [tiab] OR arterial [tiab] OR venous [tiab]) AND (prostheses [tiab] OR prosthesis [tiab] OR graft [tiab] OR grafts [tiab] OR scaffold [tiab] OR scaffolds [tiab] OR replacement [tiab] OR replacements [tiab] OR substitute [tiab] OR substitutes [tiab])))
Component 3: animal (adapted version of[1,2])	(“animal experimentation”[MeSH Terms] OR “models, animal”[MeSH Terms] OR “invertebrates”[MeSH Terms] OR “Animals”[Mesh:noexp] OR “animal population groups”[MeSH Terms] OR “chordata”[MeSH Terms:noexp] OR “chordata, nonvertebrate”[MeSH Terms] OR “vertebrates”[MeSH Terms:noexp] OR “amphibians”[MeSH Terms] OR “birds”[MeSH Terms] OR “fishes”[MeSH Terms] OR “reptiles”[MeSH Terms] OR “mammals”[MeSH Terms:noexp] OR “primates”[MeSH Terms:noexp] OR “artiodactyla”[MeSH Terms] OR “carnivora”[MeSH Terms] OR “cetacea”[MeSH Terms] OR “chiroptera”[MeSH Terms] OR “elephants”[MeSH Terms] OR “hyraxes”[MeSH Terms] OR “insectivora”[MeSH Terms] OR “lagomorpha”[MeSH Terms] OR “marsupialia”[MeSH Terms] OR “monotremata”[MeSH Terms] OR “perissodactyla”[MeSH Terms] OR “rodentia”[MeSH Terms] OR “scandentia”[MeSH Terms] OR “sirenia”[MeSH Terms] OR “xenarthra”[MeSH Terms] OR “haplorhini”[MeSH Terms:noexp] OR “strepsirhini”[MeSH Terms] OR “platyrrhini”[MeSH Terms] OR “tarsii”[MeSH Terms] OR “catarrhini”[MeSH Terms:noexp] OR “cercopithecidae”[MeSH Terms] OR “hylobatidae”[MeSH Terms] OR “hominidae”[MeSH Terms:noexp] OR “gorilla gorilla”[MeSH Terms] OR “pan paniscus”[MeSH Terms] OR “pan troglodytes”[MeSH Terms] OR “pongo pygmaeus”[MeSH Terms]) OR ((animals[tiab] OR animal[tiab] OR mice[Tiab] OR mus[Tiab] OR mouse[Tiab] OR murine[Tiab] OR woodmouse[tiab] OR rats[Tiab] OR rat[Tiab] OR murinae[Tiab] OR muridae[Tiab] OR cottonrat[tiab] OR cottonrats[tiab] OR hamster[tiab] OR hamsters[tiab] OR cricetinae[tiab] OR rodentia[Tiab] OR rodent[Tiab] OR rodents[Tiab] OR pigs[Tiab] OR pig[Tiab] OR swine[tiab] OR swines[tiab] OR piglets[tiab] OR piglet[tiab] OR boar[tiab] OR boars[tiab] OR “sus scrofa”[tiab] OR ferrets[tiab] OR ferret[tiab] OR polecat[tiab] OR polecats[tiab] OR “mustela putorius”[tiab] OR “guinea pigs”[Tiab] OR “guinea pig”[Tiab] OR cavia[Tiab] OR callithrix[Tiab] OR marmoset[Tiab] OR marmosets[Tiab] OR cebuella[Tiab] OR hapale[Tiab] OR octodon[Tiab] OR chinchilla[Tiab] OR chinchillas[Tiab] OR gerbillinae[Tiab] OR gerbil[Tiab] OR gerbils[Tiab] OR jird[Tiab] OR jirds[Tiab] OR merione[Tiab] OR meriones[Tiab] OR rabbits[Tiab] OR rabbit[Tiab] OR hares[Tiab] OR hare[Tiab] OR diptera[Tiab] OR flies[Tiab] OR fly[Tiab] OR dipteral[Tiab] OR drosophila[Tiab] OR drosophilidae[Tiab] OR cats[Tiab] OR cat[Tiab] OR carus[Tiab] OR felis[Tiab] OR nematoda[Tiab] OR nematode[Tiab] OR nematodes[Tiab] OR sipunculida[Tiab] OR dogs[Tiab] OR dog[Tiab] OR canine[Tiab] OR canines[Tiab] OR canis[Tiab] OR sheep[Tiab] OR sheeps[Tiab] OR mouflon[Tiab] OR mouflons[Tiab] OR ovis[Tiab] OR goats[Tiab] OR goat[Tiab] OR capra[Tiab] OR capras[Tiab] OR rupicapra[Tiab] OR rupicapras[Tiab] OR chamois[Tiab] OR haplorhini[Tiab] OR monkey[Tiab] OR monkeys[Tiab] OR anthropoidea[Tiab] OR anthropoids[Tiab] OR saguinus[Tiab] OR tamarin[Tiab] OR tamarins[Tiab] OR leontopithecus[Tiab] OR hominidae[Tiab] OR ape[Tiab] OR apes[Tiab] OR “pan paniscus”[Tiab] OR bonobo[Tiab] OR bonobos[Tiab] OR “pan troglodytes”[Tiab] OR gibbon[Tiab] OR gibbons[Tiab] OR siamang[Tiab] OR siamangs[Tiab] OR nomascus[Tiab] OR symphalangus[Tiab] OR chimpanzee[Tiab] OR chimpanzees[Tiab] OR prosimian[Tiab] OR prosimians[Tiab] OR “bush baby”[Tiab] OR bush babies[Tiab] OR galagos[Tiab] OR galago[Tiab] OR pongidae[Tiab] OR gorilla[Tiab] OR gorillas[Tiab] OR “pongo pygmaeus”[Tiab] OR orangutan[Tiab] OR orangutans[Tiab] OR lemur[Tiab] OR

	lemurs[Tiab] OR lemuridae[Tiab] OR horse[Tiab] OR horses[Tiab] OR equus[Tiab] OR cow[Tiab] OR calf[Tiab] OR bull[Tiab] OR chicken[Tiab] OR chickens[Tiab] OR gallus[Tiab] OR quail[Tiab] OR bird[Tiab] OR birds[Tiab] OR quails[Tiab] OR poultry[Tiab] OR poultries[Tiab] OR fowl[Tiab] OR fowls[Tiab] OR reptile[Tiab] OR reptilia[Tiab] OR reptiles[Tiab] OR snakes[Tiab] OR snake[Tiab] OR lizard[Tiab] OR lizards[Tiab] OR alligator[Tiab] OR alligators[Tiab] OR crocodile[Tiab] OR crocodiles[Tiab] OR turtle[Tiab] OR turtles[Tiab] OR amphibian[Tiab] OR amphibians[Tiab] OR amphibia[Tiab] OR frog[Tiab] OR frogs[Tiab] OR bombina[Tiab] OR salientia[Tiab] OR toad[Tiab] OR toads[Tiab] OR "epidalea calamita"[Tiab] OR salamander[Tiab] OR salamanders[Tiab] OR eel[Tiab] OR eels[Tiab] OR fish[Tiab] OR fishes[Tiab] OR pisces[Tiab] OR catfish[Tiab] OR catfishes[Tiab] OR siluriformes[Tiab] OR arius[Tiab] OR heteropneustes[Tiab] OR sheatfish[Tiab] OR perch[Tiab] OR perches[Tiab] OR percidae[Tiab] OR perca[Tiab] OR trout[Tiab] OR trouts[Tiab] OR char[Tiab] OR chars[Tiab] OR salvelinus[Tiab] OR minnow[Tiab] OR cyprinidae[Tiab] OR carps[Tiab] OR carp[Tiab] OR zebrafish[Tiab] OR zebrafishes[Tiab] OR goldfish[Tiab] OR goldfishes[Tiab] OR guppy[Tiab] OR guppies[Tiab] OR chub[Tiab] OR chubs[Tiab] OR tinca[Tiab] OR barbels[Tiab] OR barbus[Tiab] OR pimephales[Tiab] OR promelas[Tiab] OR "poecilia reticulata"[Tiab] OR mullet[Tiab] OR mullets[Tiab] OR eel[Tiab] OR eels[Tiab] OR seahorse[Tiab] OR seahorses[Tiab] OR mugil curema[Tiab] OR atlantic cod[Tiab] OR shark[Tiab] OR sharks[Tiab] OR catshark[Tiab] OR anguilla[Tiab] OR salmonid[Tiab] OR salmonids[Tiab] OR whitefish[Tiab] OR whitefishes[Tiab] OR salmon[Tiab] OR salmons[Tiab] OR sole[Tiab] OR solea[Tiab] OR lamprey[Tiab] OR lampreys[Tiab] OR pumpkinseed[Tiab] OR sunfish[Tiab] OR sunfishes[Tiab] OR tilapia[Tiab] OR tilapias[Tiab] OR turbot[Tiab] OR turbots[Tiab] OR flatfish[Tiab] OR flatfishes[Tiab] OR sciuridae[Tiab] OR squirrel[Tiab] OR squirrels[Tiab] OR chipmunk[Tiab] OR chipmunks[Tiab] OR suslik[Tiab] OR susliks[Tiab] OR vole[Tiab] OR voles[Tiab] OR lemming[Tiab] OR lemmings[Tiab] OR muskrat[Tiab] OR muskrats[Tiab] OR lemmus[Tiab] OR otter[Tiab] OR otters[Tiab] OR marten[Tiab] OR martens[Tiab] OR martes[Tiab] OR weasel[Tiab] OR badger[Tiab] OR badgers[Tiab] OR ermine[Tiab] OR mink[Tiab] OR minks[Tiab] OR sable[Tiab] OR sables[Tiab] OR gulo[Tiab] OR gulos[Tiab] OR wolverine[Tiab] OR wolverines[Tiab] OR mustela[Tiab] OR llama[Tiab] OR llamas[Tiab] OR alpaca[Tiab] OR alpacas[Tiab] OR camelid[Tiab] OR camelids[Tiab] OR guanaco[Tiab] OR guanacos[Tiab] OR chiroptera[Tiab] OR chiropteras[Tiab] OR bat[Tiab] OR bats[Tiab] OR fox[Tiab] OR foxes[Tiab] OR iguana[Tiab] OR iguanas[Tiab] OR xenopus laevis[Tiab] OR parakeet[Tiab] OR parakeets[Tiab] OR parrot[Tiab] OR parrots[Tiab] OR donkey[Tiab] OR donkeys[Tiab] OR mule[Tiab] OR mules[Tiab] OR zebra[Tiab] OR zebras[Tiab] OR shrew[Tiab] OR shrews[Tiab] OR bison[Tiab] OR bisons[Tiab] OR buffalo[Tiab] OR buffaloes[Tiab] OR deer[Tiab] OR deers[Tiab] OR bear[Tiab] OR bears[Tiab] OR panda[Tiab] OR pandas[Tiab] OR "wild hog"[Tiab] OR "wild boar"[Tiab] OR fitchew[Tiab] OR fitch[Tiab] OR beaver[Tiab] OR beavers[Tiab] OR jerboa[Tiab] OR jerboas[Tiab] OR capybara[Tiab] OR capybaras[Tiab] OR canine [tiab] OR bovine [tiab] OR porcine [tiab] OR hog [tiab] OR hogs [tiab]) OR baboon[tiab] OR baboons[tiab] OR non-human primate[tiab] OR non-human primates[tiab] OR non human primate[tiab] OR non human primates [tiab] OR monkey[tiab] OR monkeys[tiab]

Supplementary Table 2: Search strategy for Embase using search string defining the components “blood vessel prosthesis”, “tissue engineering” and “animal”

Component	Search entry
Component 1: blood vessel prosthesis	exp Blood Vessel Prosthesis/ or (((artery or arteries or vein or veins or vascular or arterial or venous) and (prostheses or prosthesis or graft or grafts or scaffold or scaffolds or replacement or replacements or substitute or substitutes or grafting))
Component 2: tissue engineering	ti,ab,kw. and ((tissue engineering or tissue engineered).ti,ab,kw. or exp Tissue Engineering/ or tissue regeneration/ or exp tissue repair/ or guided Tissue Regeneration.ti,ab,kw. or regeneration.ti,ab,kw.)) or TEVG.ti,ab,kw. or ((artery or arteries or vein or veins or vascular or arterial or venous).ti,ab,kw. and ((tissue engineering or tissue engineered).ti,ab,kw. or exp Tissue Engineering/ or tissue regeneration/ or exp tissue repair/ or guided Tissue Regeneration.ti,ab,kw.)) or Tissue engineered blood vessel.ti,ab,kw. or Tissue engineered blood vessels.ti,ab,kw. or Blood vessel tissue engineering.ti,ab,kw. or Blood vessel regeneration.ti,ab,kw. or ((degradable or biodegradable or resorbable or bioresorbable or regenerative) and (blood vessel or artery or arteries or vein or veins or vascular or arterial or venous) and (prostheses or prosthesis or graft or grafts or scaffold or scaffolds or replacement or replacements or substitute or substitutes)).ti,ab,kw.
Component 3: animal (adapted version of[1,2])	exp animal experiment/ or exp animal model/ or exp experimental animal/ or exp transgenic animal/ or exp male animal/ or exp female animal/ or exp juvenile animal/ or animal/ or chordata/ or vertebrate/ or tetrapod/ or exp fish/ or amniote/ or exp amphibia/ or mammal/ or exp reptile/ or exp sauropsid/ or therian/ or exp monotremate/ or placental mammals/ or exp marsupial/ or Euarchontoglires/ or exp Afrotheria/ or exp Boreoeutheria/ or exp Laurasiatheria/ or exp Xenarthra/ or primate/ or exp Dermoptera/ or exp Glires/ or exp Scandentia/ or Haplorhini/ or exp prosimian/ or simian/ or exp tarsiiform/ or Catarrhini/ or exp Platyrrhini/ or ape/ or exp Cercopithecidae/ or hominid/ or exp hylobatidae/ or exp chimpanzee/ or exp gorilla/ or exp orang utan/(animal or animals or pisces or fish or fishes or catfish or catfishes or sheatfish or silurus or arius or heteropneustes or clarias or gariepinus or fathead minnow or fathead minnows or pimephales or promelas or cichlidae or trout or trouts or char or chars or salvelinus or salmo or oncorhynchus or guppy or guppies or millionfish or poecilia or goldfish or goldfishes or carassius or auratus or mullet or mullets or mugil or curema or shark or sharks or cod or cods or gadus or morhua or carp or carps or cyprinus or carpio or killifish or eel or eels or anguilla or zander or sander or lucioperca or stizostedion or turbot or turbots or psetta or flatfish or flatfishes or plaice or pleuronectes or platessa or tilapia or tilapias or oreochromis or sarotherodon or common sole or dover sole or solea or zebrafish or zebrafishes or danio or rerio or seabass or dicentrarchus or labrax or morone or lamprey).ti,ab. (lampreys or petromyzon or pumpkinseed or pumpkinseeds or lepomis or gibbosus or herring or clupea or harengus or amphibia or amphibian or amphibians or anura or salientia or frog or frogs or rana or toad or toads or bufo or xenopus or laevis or bombina or epidalea or calamita or salamander or salamanders or newt or newts or triturus or reptilia or reptile or reptiles or bearded dragon or pogona or vitticeps or iguana or iguanas or lizard or lizards or anguis fragilis or turtle or turtles or snakes or snake or aves or bird or birds or quail or quails or coturnix or bobwhite or colinus or virginianus or poultry or poultries or fowl or fowls or chicken or chickens or gallus or zebra finch or taeniopygia or guttata or canary or canaries or serinus or canaria or parakeet or parakeets or grasskeet or parrot or parrots or psittacine or psittacines or shelduck or tadorna or goose or geese or branta or leucopsis or woodlark or lullula or flycatcher or ficedula or hypoleuca or dove or doves or geopelia or cuneata or duck or ducks or greylag or graylag or anser or harrier or circus pygargus or red knot or great knot or calidris or canutus).ti,ab. (godwit or limosa or lapponica or meleagris or gallopavo or jackdaw or corvus or monedula or ruff or philomachus or pugnax or lapwing or peewit or plover or vanellus or swan or cygnus or columbianus or bewickii or gull or chroicocephalus or ridibundus or albifrons or great tit or parus or aythya or fuligula or streptopelia or risoria or spoonbill or platalea or leucorodia or blackbird or turdus or merula or blue tit or cyanistes or pigeon or pigeons or columba or pintail or anas or starling or sturnus or owl or athene noctua or pochard or ferina or cockatiel or nymphicus or hollandicus or skylark or alauda or tern or sterna or teal or crecca or oystercatcher or haematopus or ostralegus or shrew or shrews or sorex or araneus or crocidura or russula or european mole or talpa or chiroptera or bat or bats or eptesicus or serotinus or myotis or dasycneme or daubentonii or pipistrelle or pipistrellus or cat or cats or felis or catus or feline or dog or dogs or canis or canine or canines or otter or otters or lutra or badger or badgers or meles or fitchew or fitch or foumart or foulmart or ferrets or ferret or polecat or polecats or mustela or putorius or weasel or weasels or fox or foxes or vulpes or common seal or phoca or vitulina or grey

	seal or halichoerus or horse or horses or equus or equine or equidae or donkey or donkeys or mule or mules or pig or pigs or swine or swines or hog or hogs or boar or boars or porcine or piglet or piglets or sus or scrofa or llama or llamas or lama or glama or deer or deers or cervus or elaphus or cow or cows or bos taurus or bos indicus or bovine or bull or bulls or cattle or bison or bisons or sheep or sheeps or ovis aries or ovine or lamb or lambs or mouflon or mouflons or goat or goats or capra or caprine or chamois or rupicapra or leporidae).ti,ab. (lagomorpha or lagomorph or rabbit or rabbits or oryctolagus or cuniculus or laprine or hares or lepus or rodentia or rodent or rodents or murinae or mouse or mice or mus or musculus or murine or woodmouse or apodemus or rat or rats or rattus or norvegicus or guinea pig or guinea pigs or cavia or porcellus or hamster or hamsters or mesocricetus or cricetulus or cricetus or gerbil or gerbils or jird or jirds or meriones or unguiculatus or jerboa or jerboas or jaculus or chinchilla or chinchillas or beaver or beavers or castor fiber or castor canadensis or sciuridae or squirrel or squirrels or sciurus or chipmunk or chipmunks or marmot or marmots or marmota or suslik or susliks or spermophilus or cynomys or cottonrat or cottonrats or sigmodon or vole or voles or microtus or myodes or glareolus or primate or primates or prosimian or prosimians or lemur or lemurs or lemuridae or loris or bush baby or bush babies or bushbaby or bushbabies or galago or galagos or anthropoidea or anthropoids or simian or simians or monkey or monkeys or marmoset or marmosets or callithrix or cebuella or tamarin or tamarins or saguinus or leontopithecus or squirrel monkey or squirrel monkeys or saimiri or night monkey or night monkeys or owl monkey or owl monkeys or douroucoulis or aotus or spider monkey or spider monkeys or ateles or baboon or baboons or papio or rhesus monkey or macaque or macaca or mulatta or cynomolgus or fascicularis or green monkey or green monkeys or chlorocebus or vervet or vervets or pygerythrus or hominoidea or ape or apes or hylobatidae or gibbon or gibbons or siamang or siamangs or nomascus or symphalangus or hominidae or orangutan or orangutans or pongo or chimpanzee or chimpanzees or pan troglodytes or bonobo or bonobos or pan paniscus or gorilla or gorillas or troglodytes).ti,ab. (non-human primate or non-human primates or non human primate or non human primates).ti,ab.

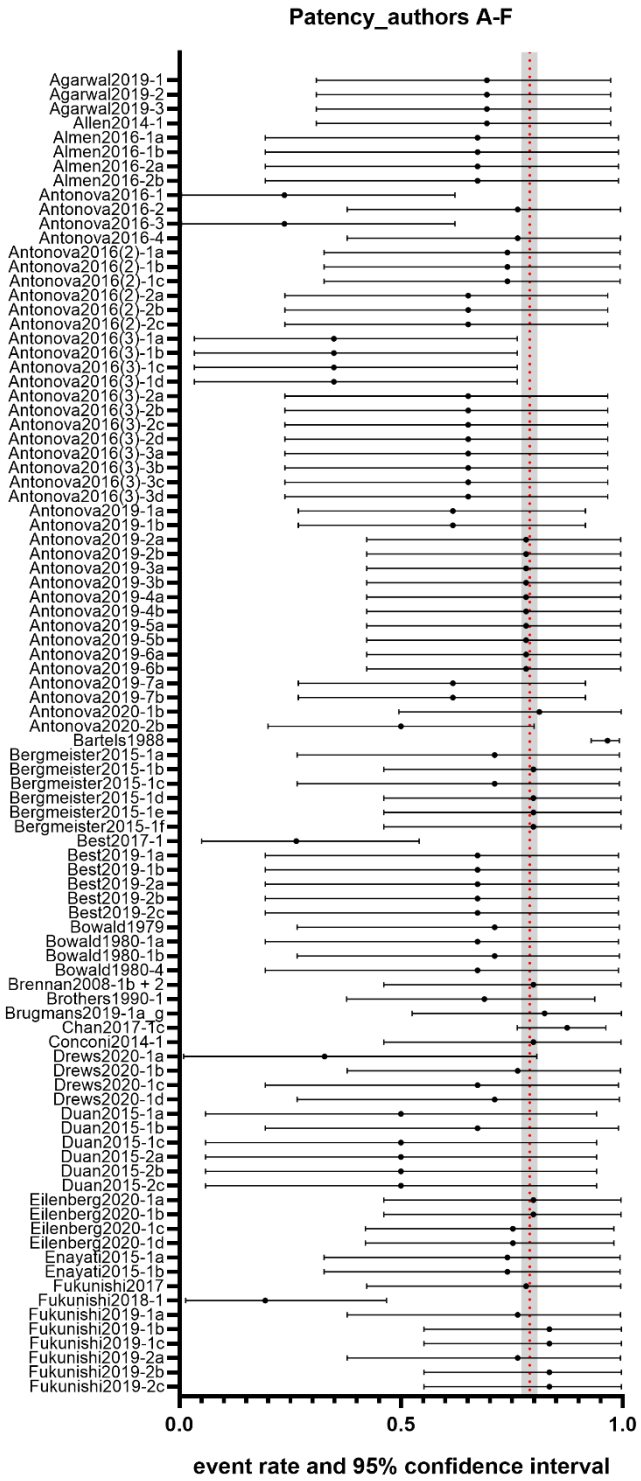
Supplementary Table 3: abbreviations of polymers

Abbreviation	Full name	Examples within this category and commercial names
CE-UPY	Chain extended- ureido-pyrimidinone	
P(BT-BLOCK-EG)	poly(polybutylene terephthalate-co-polyether glycol)	Hyrtel
P(LA-CO-GA)	poly(lactid acid-co-glycolic acid)	PHO-PGA
P(LLA-CO-CL)	poly(L-Lactic-co-caprolactone)	PLCL, PHBV/PCL, PCL2000-U4U, PPF
PA	polyamide	Nylon
PBF	poly (2, 3-butylene furmarate	
PCL	polycaprolactone	hPCL tPCL
PCU	polycarbonate urethane	dPCU(BHPC_211), reinforced Corvita
PDLA	poly-D-L-lactic acid	
PDMS	polydimethylsiloxane	
PDS	polydioxanone	
PEG	polyethylene glycol	
PETU	poly(ether)urethane	Estane
PEU	polyesterurethane	Pulsetec, poly(1-LEU-10)
PEUU	polyesterurethane urea	Mitrathane (PEUU), Thoratec
PGA	polyglycolic acid	PG10, PG910, Vicryl
PHEA	a,b-poly(N-2-hydroxyethyl)-D,L-aspartamide	
PLLA	poly-l-lactic acid	
PMEH	poly(2-methacryloyloxyethyl phosphorylcholine-co-2-Ethylhexyl methacrylate)	
PTX	Paclitaxel	
PU	polyurethane	Thermoplastic polyurethanes, segmented polyurethane (SPU), Biomer, Mitrathane (PU)
PUSN	poly(ester urethane)urea with disulfide and amino group	
PVA	polyvinyl alcohol	Ivalon sponge
SE	synthetic elastin	

Supplementary Table 4: definition of answers for quality assessment of study design and reporting of preclinical studies.

Experimental set-up	
Q1 Were there different experimental groups? Yes → continue next 2 questions. No → stop	1 experimental group = same material, same implant side, multiple time points explanted No means that there was only 1 experimental group.
Q2 Was allocation of animals to experimental groups random?	Yes: when stated in the text that they were randomly assessed, or that the grafts were randomly implanted / surgeon was blinded. No: when stated that that was not the case, or because of experimental design it was not random. Unclear: when it was not described. N.A.: when there were no different experimental groups
Q3 Was the outcome assessor blinded for the experimental groups (for any compared qualitative outcome?)	N.A.: when there were no different experimental groups Unclear: when it was not stated if the assessor was blinded. Yes: when it was stated that the outcome assessor was blinded. No: when it was stated that the outcome assessor was not blinded
Study Outcome	
Q4 Are multiple locations within the vascular graft analyzed per animal?	Yes: when observations were compared between proximal, distal, mid-graft or anastomosis side. Or when angiogram, doppler ultra sound or complete full length staining/SEM analysis was performed No: when it was stated in method section that only the mid portion was analyzed Unclear: when it was not stated, nor became clear from M&M
Q5 Are explanted vascular graft analyzed randomly selected?	Yes when it was stated that the outcome assessor was blinded, or when the order of analysis of the samples was random. Unclear: when it was not stated if the samples were randomly assessed. NA when there was only 1 included animal.
Reporting of Animal Information	
Q6 Is the animal species described?	Yes: animal species described
Q7 Is the animal strain described?	Unclear: when it was only described for some experimental groups
Q8 Is the sex of animals described?	Unclear: when it was only described for some experimental groups
Q9 Is the age of animals described?	Unclear when only “young” or “adult” “juvenile” was given
Q10 Is the weight of animals described?	
Q11 Is the ethical review board described?	Yes: when board was given, but also when referred to NIH guidelines Unclear: when only stated : “approved by a local ethic committee”
Q12 Is the number of animals allocated to each experimental group described?	Unclear: when not described in material and methods, and only mentioned in 1 figure legend
Q13 Is the allocated follow-up time described?	Unclear: when not described for all experimental groups
Reporting of Procedure:	
Q14 Is the synthetic material of the vascular graft described?	Yes: this was an inclusion criterium
Q15 Are the dimensions of the vascular graft described? (length, diameter, wall thickness)	Yes: when all 3 values were given: length, inner diameter and wall thickness Unclear: when only 1/3 or 2/3 dimensions were given No: when none of these were given

Q16 Is sterilization/disinfection of the vascular grafts described?	Unclear: when not described for all experimental groups / or when only mentioned "worked sterile"/ "sterilized" but no method given
Q17 Is the vascular graft storage prior to implantation described?	Unclear: when only stated "packed"
Q18 If applicable, is the procedure for functionalization of the vascular graft described?	Unclear: when it is described, but no clear method is given and no references to other papers in which it is described
Q19 If applicable, is the procedure for on-the-fly cell seeding of the vascular graft described?	Yes: described or referred to a source describing the seeding process.
Q20 Is the location of implantation of the vascular graft described?	Yes: all answered yes, because one of the inclusion criteria
Q21 Is the surgical procedure of implantation of the vascular graft described (including references to other papers)?	No: when no explanation is given and no references to other papers Unclear when no reference is given and when the surgical procedure is hardly described
Q23 Is the use and period of anti-coagulation described?	Unclear: when only the use is given during operation or only post implantation, but not both No: not at all stated if and when anticoagulation was used. Could also be the case for when authors refer to other source for the implantation procedure, but then not "briefly" describe the most important details like eg use of anticoagulation.
Q23 Is the use and period of anaesthesia, analgesia described?	Unclear when only was stated "animals were anesthetized" No: when not reported. Could also be the case for when authors refer to other source for the implantation procedure, but then not "briefly" describe the most important details like eg anesthesia.
Reporting of general outcome:	
Q24 Are adverse events described?	Unclear: only based on occlusion rates, but not really described that they looked for eg thrombosis / stenosis / aneurysm / other complications
Q25 Is the final number of animals analyzed in each experimental group per timepoint described?	Yes: when the initial number of included animals was clear and the drop outs per group. No: when the initial number of included animals were not clear Unclear: when the drop outs per timepoint were not clear, but for the overall group were clear
Q26 Are numbers of drop-outs described?	Unclear: when not given for all experimental groups. Or when not clearly stated, but only when calculating by the reader the drop outs can be known. NA: when only 1 animal was included and it survived
Q26a If applicable: Are the reasons for drop-outs described?	Yes: when cause was given, NB: also when stated "death due to unknown cause" No: when not given, only mentioned death. NA when all animals survived and no animals dropped out Unclear when animals dropped out, and you know the number, but not the reason why
Q26b If applicable: Is the timepoint of drop-outs described?	Yes: timepoint is mentioned (perioperative or the timeframe in which the animals died prior to planned explant-time) No: when not given, only mentioned deaths per experimental groups. NA when all animals survived and no animals dropped out Unclear when animals dropped out, and you know the amount, but not the timepoint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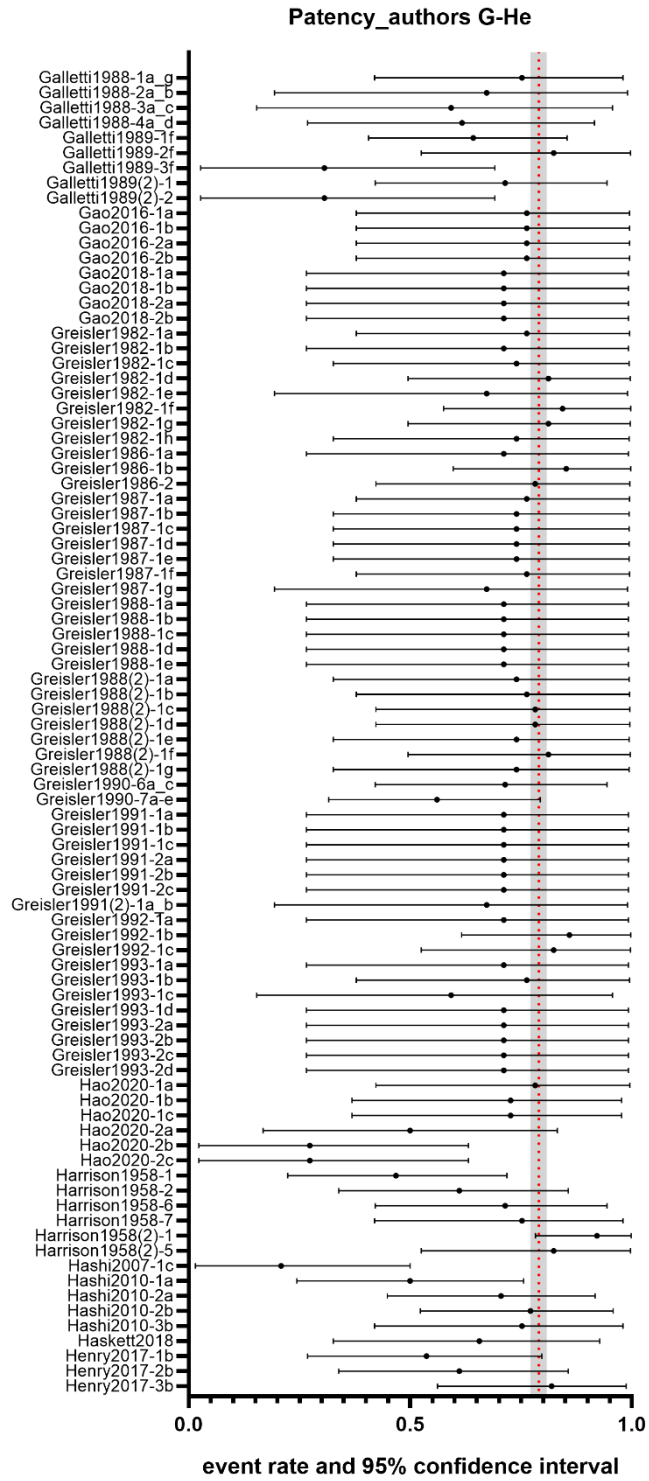
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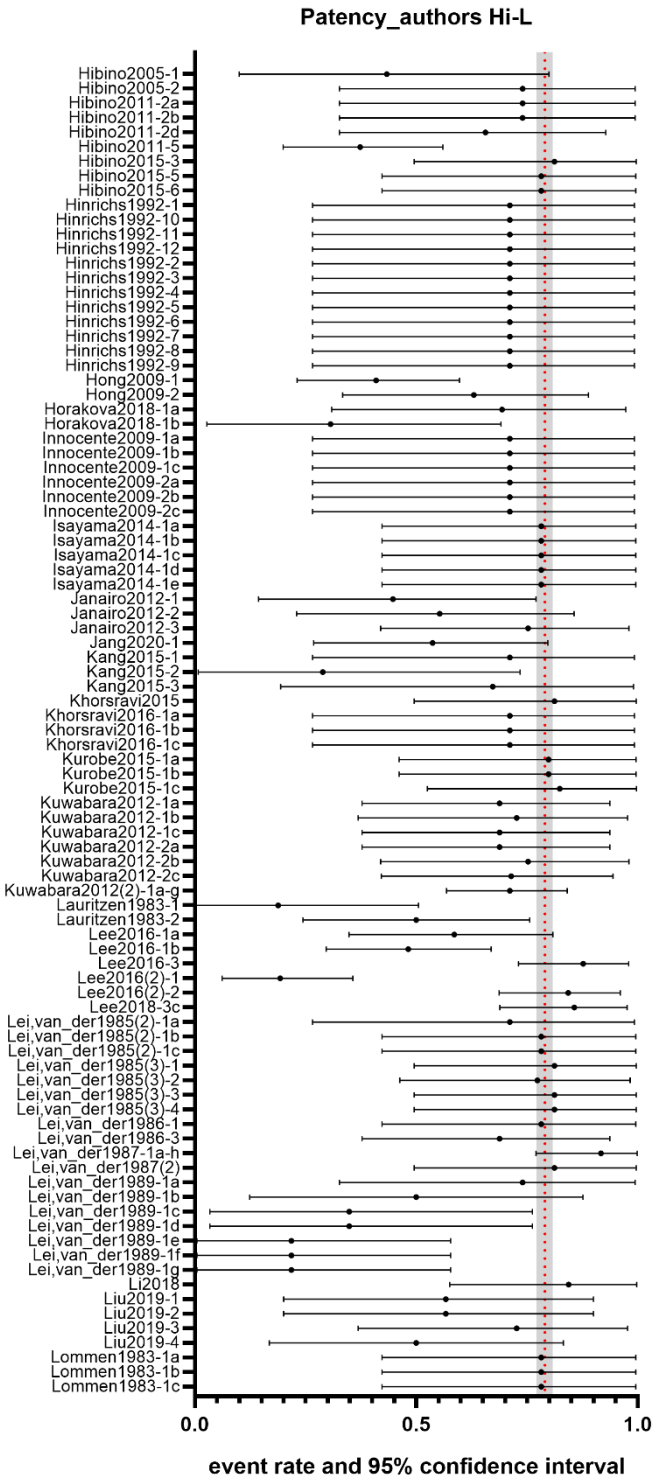
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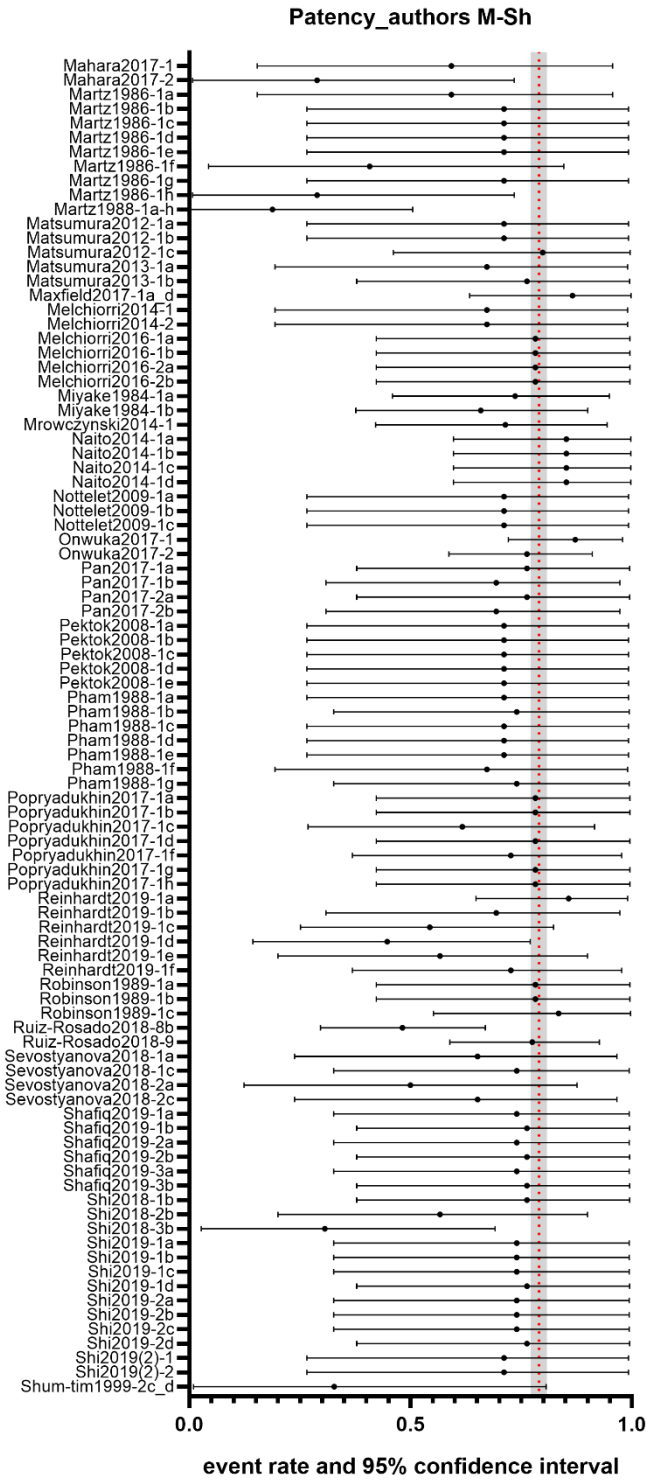
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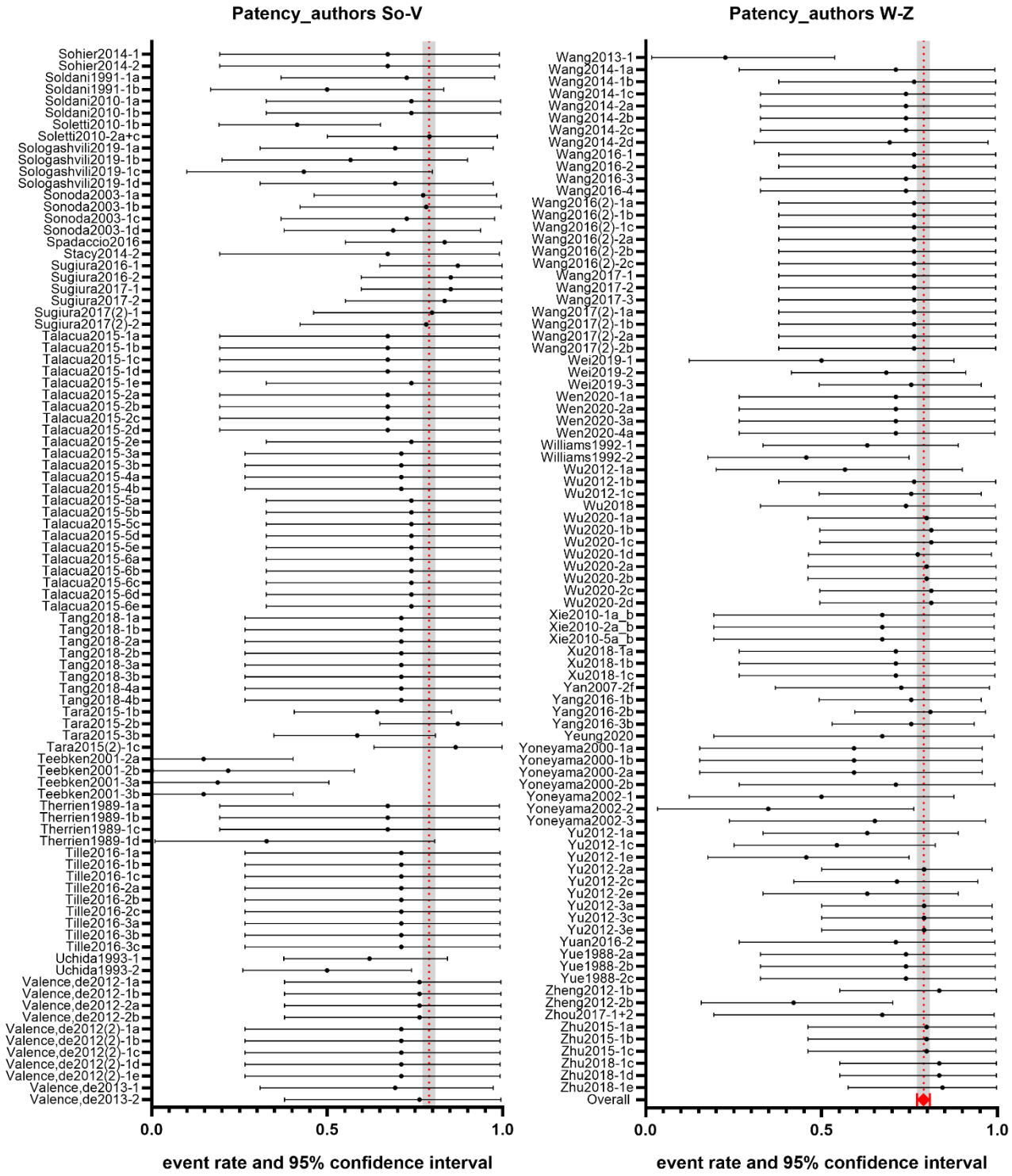
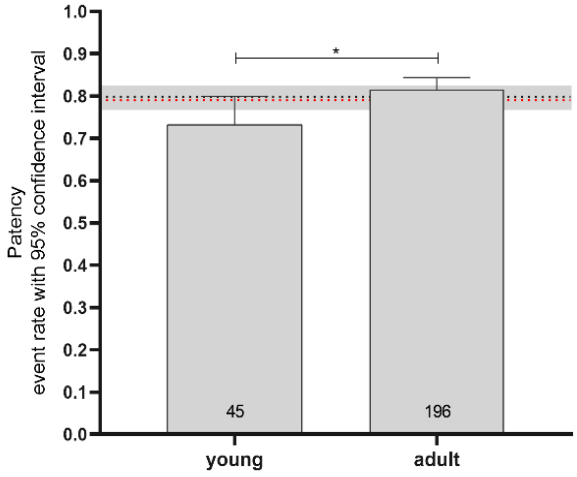


Supplementary Figure 1 Forest plot of overall patency rate of all included experimental groups illustrated as event rate with 95% conficence interval.

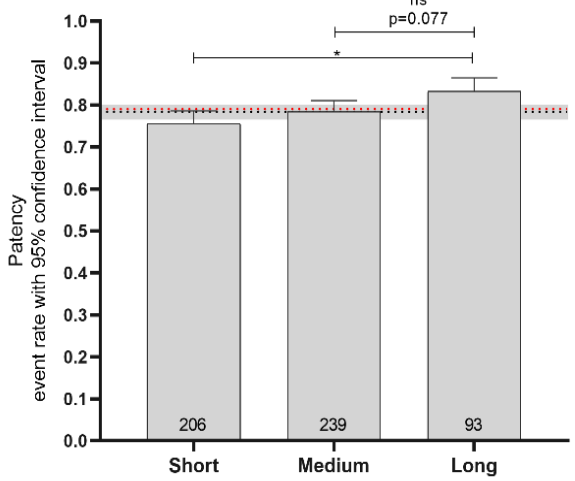
a.

	Age puberty (average species)	Cut-off value sensitivity analysis young vs adult	Total life expectancy (average species)	Cut-off value sensitivity analysis short vs medium follow up (1.67% of total life expectancy)	Cut-off value sensitivity analysis medium vs long follow up (6.67% of total life expectancy)	References
Mouse	5-7w	7w	2y	14d	60d	[3,4]
Rat	7-11w	11w	3y	21d	90d	[4,5]
Rabbit	3-6m	6m	10y	70d	300d	[4,6,7]
Pig	4-6m	6m	15y	105d	450d	[4,8]
Dog	5-7m	7m	14y	98d	420d	[4,8]
Sheep	n.a.*		12.5y	87d	375d	[4,8]

b.

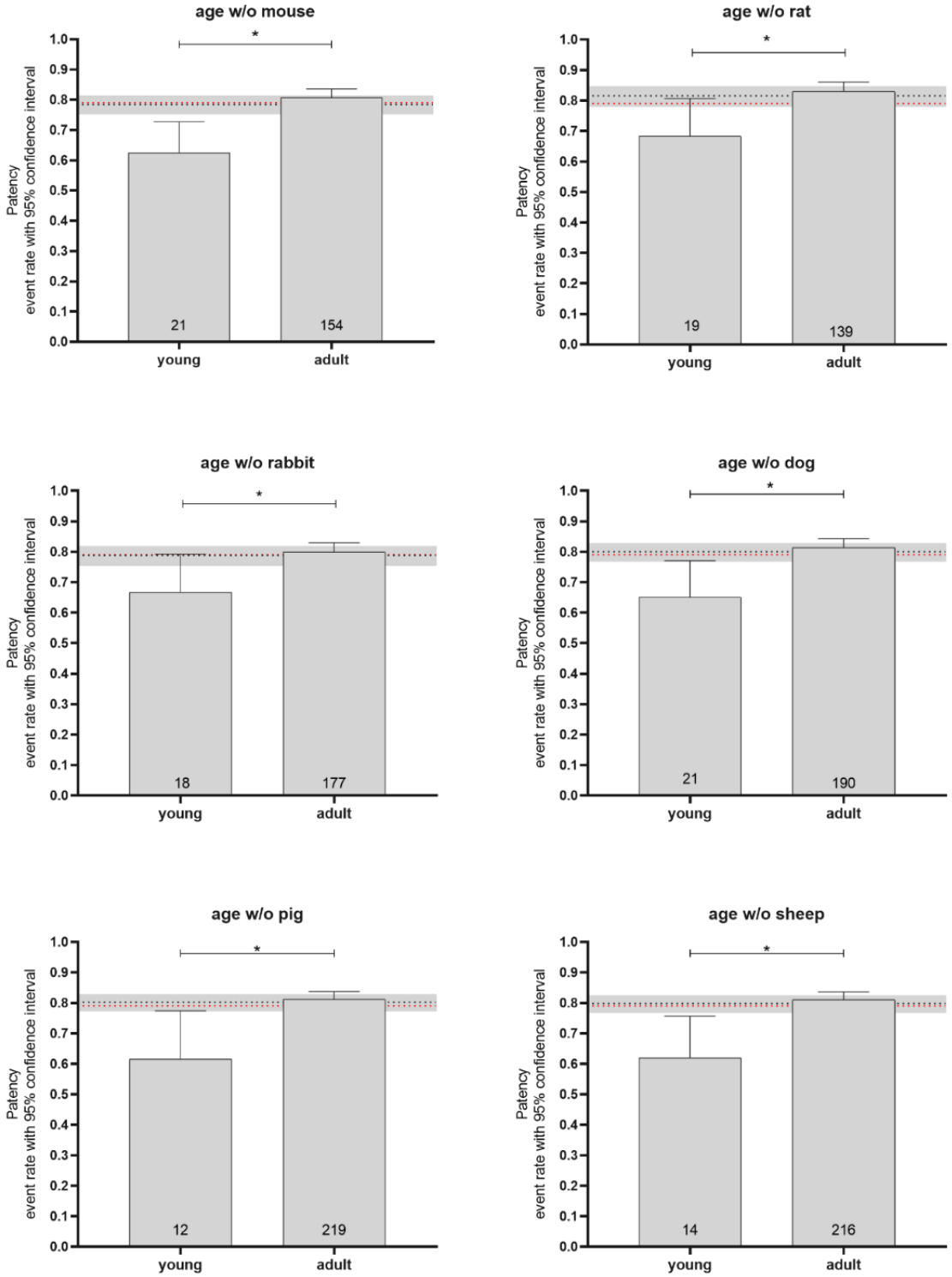


c.

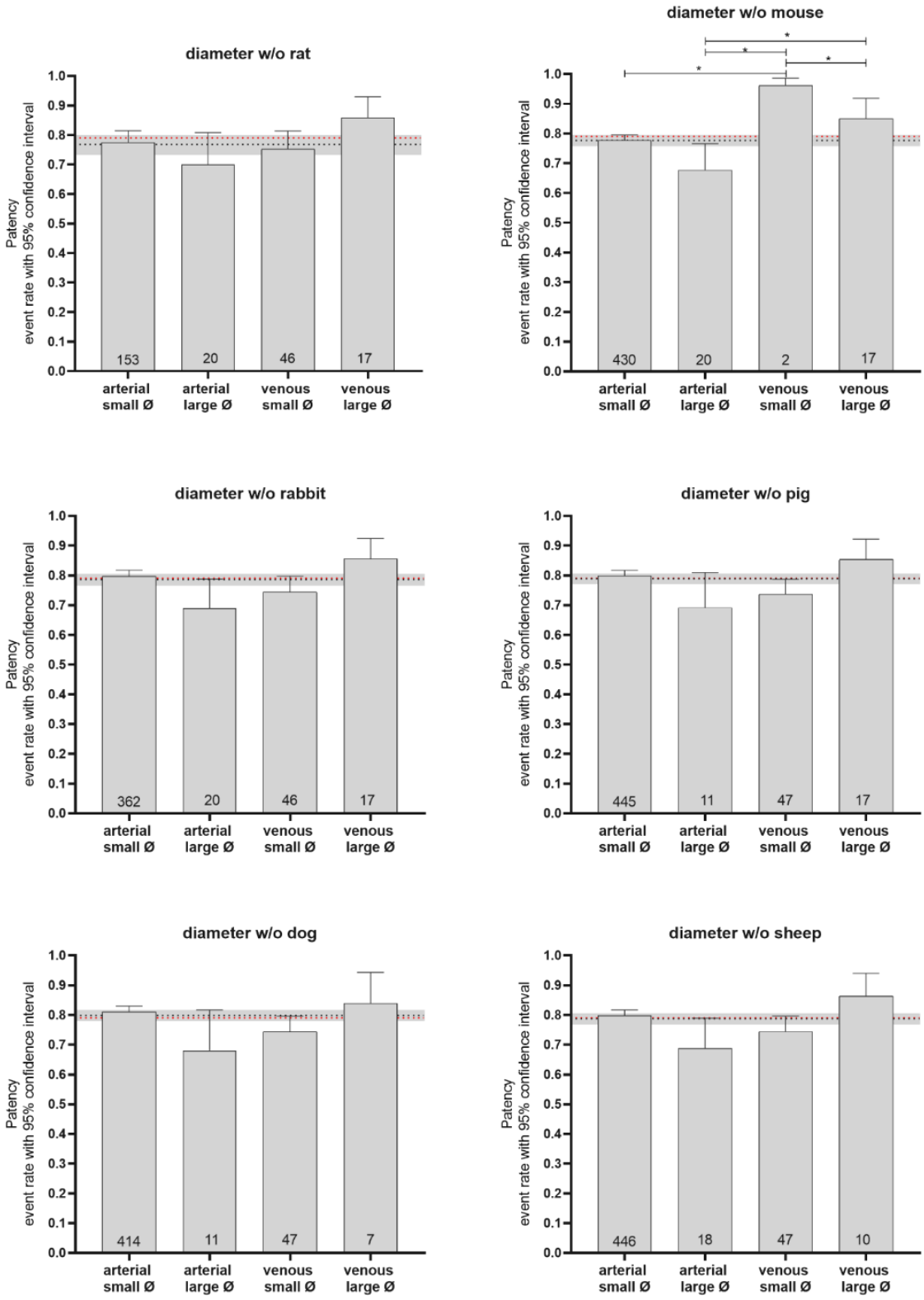


*for sheep studies: method description mentioning "juvenile" or "lamb" was sufficient to make the categorization

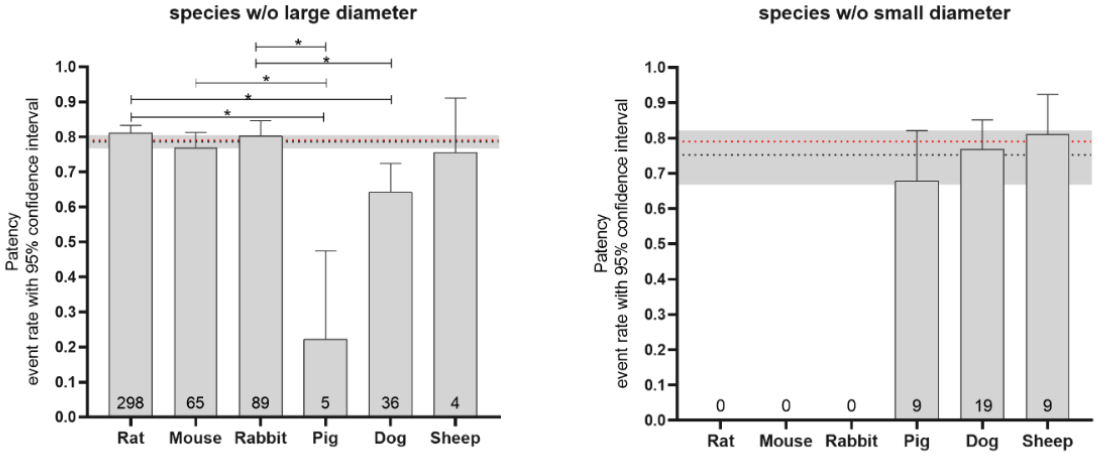
Supplementary Figure 2: Sensitivity analysis on subcategorization of age and follow-up time. A) Adjusted cut-off values for subcategorization for sensitivity analysis. B) Sensitivity analysis on age, including period of puberty with young age group, showed no differences compared to previous categorization. C) Sensitivity analysis on follow-up time showed similar trends with previous categorization. Numbers in bar represent number of experimental groups. Black dotted line and grey shades representing grouped ER with 95% CI respectively. Red dotted line: overall ER patency. Significance * $p<0.05$.



Supplementary Figure 3: Sensitivity analysis of species, showing patency rate per age categorization with exclusion of one species per graph. Numbers in bar represent number of experimental groups. Black dotted line and grey shades representing grouped ER with 95% CI respectively. Red dotted line: overall ER patency. Significance * $p<0.05$.



Supplementary Figure 4: Sensitivity analysis of species, showing patency rate per categorization for implant site and graft diameter with exclusion of one species per graph. Numbers in bar represent number of experimental groups. Black dotted line and grey shades representing grouped ER with 95% CI respectively. Red dotted line: overall ER patency. Significance * $p<0.05$



Supplementary Figure 5: Sensitivity analysis of small and large diameter categorization, showing overall patency rate per species for only small diameter grafts (left) or only large diameter grafts (right). Numbers in bar represent number of experimental groups. Black dotted line and grey shades representing grouped ER with 95% CI respectively. Red dotted line: overall ER patency. Significance * $p<0.05$.

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